

Project progress report on

Execution-Driven MIPS-Lite Pipeline Simulator

-By

Team 11

Sathwik Bandi

Bhanu Sai Sidhardha Tummala

Uma Mahesh Bestha

Thanmai Neelampalli

Current State:

- The team has assigned roles and responsibilities, but coding is not fully completed.
- The team members are familiarizing themselves with MIPS architecture, the simulator's design, and their assigned components.

Roles and Responsibilities:

Our team has clearly defined roles and responsibilities to ensure efficient progress on the project:

Sathwik Bandi:

- Responsible for implementing Logical Instructions and creating trace files for testing and debugging.
- Read the memory image trace file (hex format) line by line.
- Convert each line from hex to binary and pass it to the Instruction Decoder.
- Will work on refining the Timing Simulator.

Bhanu Sai Sidhardha Tummala:

- Will focus on decoding each instruction (R-type, I-type).
- Categorize instructions (Arithmetic, Logical, Memory Access, Control Flow).

Uma Mahesh Bestha:

- Develop and validate a timing simulator for both non-forwarding and forwarding pipelines.
- Responsible for understanding and implementing Register operations and Instruction Set Architecture.

Thanmai Neelampalli:

- Will handle the implementation of Control and Memory Instructions.
- Responsible for developing the Functional Simulator.
- Simulate the machine state (Program Counter, General Purpose Registers, Memory).
- Maintain and update the state until a "HALT" instruction is encountered.
- Create test cases to validate the functional and timing simulators.

Future Work:

- Collaboratively design and implement the complete simulator code.
- Conduct extensive debugging to ensure functionality.
- Develop a clear understanding of the MIPS pipeline architecture to optimize the simulator.

Schedule:

- Design Phase: Immediate, expected to be completed in a week
- Development Phase: Mid-May to End of May
Complete the implementation of individual components
- Testing Phase: Early June
Develop and run test cases for both Functional and Timing Simulators.
- Final Submission: June 6.

Challenges/Questions:

- Currently, there are no specific challenges reported. We would reach out to either Professor or TA, grader in case of any challenges in future.